if it is chosen by the majority of focal diversity metrics, i.e., at least three focal diversity metrics, via hierarchical pruning. This third step further refines the focal diversity powered hierarchical pruning results and consistently delivers enhanced precision and robustness in ensemble pruning.

4 FORMAL ANALYSIS

Deep neural network ensembles use multiple deep neural networks to form a team to collaborate and combine the predictions of individual member networks to make the final prediction. It is a commonly held view that an ensemble team consisting of diverse member networks has high prediction performance [18, 24, 32]. Our focal diversity based ensemble pruning methods can efficiently identify small pruned deep ensembles with highly diverse member networks. These pruned deep ensembles can achieve the same or even improved prediction performance with significantly reduced space and time costs compared to the entire deep ensemble. In this section, we present a formal analysis to show the desired properties and features of these pruned deep ensembles to further demonstrate the effectiveness of our proposed methods.

4.1 Diversity by Uncorrelated Errors

For an ensemble team of size S, following [33, 39], we can derive the added error for its ensemble prediction E_{add}^{avg} using model averaging (avg) as the ensemble consensus method as Formula (1) shows.

$$E_{add}^{avg} = E_{add}(\frac{1 + (S-1)\delta)}{S}) \tag{1}$$

where E_{add} is the added error of a single network and δ is the expected average correlation of all member networks in the ensemble. Therefore, the ideal scenario is when all member networks in an ensemble team of size S are diverse. They can learn and predict with uncorrelated errors (failure independence), i.e., $\delta \leq 0$. Then a simple model averaging method can significantly reduce the overall prediction error by at least S times. Meanwhile, the worst scenario happens when errors of individual networks are highly correlated with $\delta = 1$. For example, when all S member networks are perfect duplicates, the error of the ensemble is identical to the initial error without any improvement. In practice, given that it is challenging to directly measure the correlation δ, many ensemble diversity metrics are proposed to quantify the correlation among member networks of an ensemble team. Our focal diversity metrics can significantly improve ensemble diversity measurement, and they are closely correlated to the ensemble prediction performance, which can be directly leveraged for identifying high-quality ensemble teams.

4.2 Ensemble Robustness

The deep neural network model is typically trained to minimize a cross-entropy loss and output a probability vector to approximate posteriori probabilities for the corresponding classes. Let $f_i(\mathbf{x})$ denote the ith element in the probability vector of a classifier F for input $\mathbf{x}$, which predicts the probability of class i. The classifier F will output the predicted label c with the highest probability for input $\mathbf{x}$, that is $c = argmax_{1 \leq i \leq C} f_i(\mathbf{x})$.

The robustness of a classifier F can be assessed based on its ability to maintain consistent prediction for a given input $\mathbf{x}$ under input perturbation (μ), such as the noise from adversarial samples. When the magnitude of the input perturbation μ remains within a certain bound, the prediction made by the classifier F remains unaffected. We can leverage such a bound to compare the level of robustness of different classifiers, where a higher bound indicates a higher level of robustness. In Theorem 1, we introduce the concept of robustness bound (R) and formally show that this bound fulfills the aforementioned requirement.

Theorem 1 (Robustness Bound (R))**.** The robustness bound (R) for a classifier F can be defined as the following Formula (2),

$$R = min_{j \neq c} \frac{f_c(\mathbf{x}_0) - f_j(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(f_c(\mathbf{x}) - f_j(\mathbf{x}))||_q} \tag{2}$$

where $\mathbf{x} = \mathbf{x}_0 + \mu$. When the magnitude of the perturbation μ on the input $\mathbf{x}_0$ is limited by $||\mu||_p \leq R$ and p, q satisfy $\frac{1}{p} + \frac{1}{q} = 1$ and $1 \leq p, q \leq \infty$, the predicted labels for $\mathbf{x}$ and $\mathbf{x}_0$ by the classifier F will be identical, indicating the input perturbation (μ) will not change the prediction.

We first introduce Lemma 1 on Lipschitz continuity below and then show the formal proof of Theorem 1.

Lemma 1 (Lipschitz Continuity)**.** If $g(\mathbf{x})$ is Lipschitz continuous, according to [28], the following inequality (3) holds:

$$|g(\mathbf{x}) - g(\mathbf{y})| \leq L_q ||\mathbf{x} - \mathbf{y}||_p \tag{3}$$

where $\mathbf{x}$ and $\mathbf{y}$ are inputs, Lipschitz constant $L_q = max_{\mathbf{x}} ||\nabla g(\mathbf{x})||_q$, $\frac{1}{p} + \frac{1}{q} = 1$, and $1 \leq p, q \leq \infty$.

Proof of Theorem 1. Without loss of generality, we assume $g(\mathbf{x}) = f_c(\mathbf{x}) - f_j(\mathbf{x})$ is Lipschitz continuous with Lipschitz constant L_q^j and $j \neq c$. Following Lemma 1, let $\mathbf{x} = \mathbf{x}_0 + \mu$ and $\mathbf{y} = \mathbf{x}_0$, we derive Formula (4) below,

$$|g(\mathbf{x}_0 + \mu) - g(\mathbf{x}_0)| \leq L_q^j ||\mu||_p \tag{4}$$

where $L_q^j = max_{\mathbf{x}} ||\nabla g(\mathbf{x})||_q = max_{\mathbf{x}} ||\nabla(f_c(\mathbf{x}) - f_j(\mathbf{x}))||_q$, $\frac{1}{p} + \frac{1}{q} = 1$, and $1 \leq p, q \leq \infty$.

Formula (4) can be rearranged as Formula (5),

$$g(\mathbf{x}_0) - L_q^j ||\mu||_p \leq g(\mathbf{x}_0 + \mu) \leq g(\mathbf{x}_0) + L_q^j ||\mu||_p \tag{5}$$

when $g(\mathbf{x}_0 + \mu) < 0$, the predicted class label will change. However, $g(\mathbf{x}_0 + \mu)$ is lower bounded by $g(\mathbf{x}_0) - L_q^j ||\mu||_p \leq g(\mathbf{x}_0 + \mu)$. If $0 \leq g(\mathbf{x}_0) - L_q^j ||\mu||_p$, we have $g(\mathbf{x}_0 + \mu) \geq 0$ to ensure that the prediction will not change with the small perturbation μ on the input $\mathbf{x}_0$. This leads to Formula (6),

$$g(\mathbf{x}_0) - L_q^j ||\mu||_p \geq 0 \Rightarrow ||\mu||_p \leq \frac{g(\mathbf{x}_0)}{L_q^j} \tag{6}$$

that is Formula (7):

$$||\mu||_p \leq \frac{f_c(\mathbf{x}_0) - f_j(\mathbf{x}_0)}{L_q^j} \tag{7}$$

where $L_q^j = max_{\mathbf{x}} ||\nabla(f_c(\mathbf{x}) - f_j(\mathbf{x}))||_q$, $\frac{1}{p} + \frac{1}{q} = 1$, and $1 \leq p, q \leq \infty$.

In order to ensure the classification result will not change, that is $argmax_{1 \leq i \leq C} f_i(\mathbf{x}_0 + \mu) = c$, we use the minimum of the bound on μ over $j \neq c$ to obtain the inequality (8) with $\frac{1}{p} + \frac{1}{q} = 1$ and $1 \leq p, q \leq \infty$,

$$||\mu||_p \leq min_{j \neq c} \frac{f_c(\mathbf{x}_0) - f_j(\mathbf{x}_0)}{L_q^j} = min_{j \neq c} \frac{f_c(\mathbf{x}_0) - f_j(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(f_c(\mathbf{x}) - f_j(\mathbf{x}))||_q} = R \tag{8}$$

which indicates that as long as $||\mu||_p$ is small enough to fulfill the above bound, the classifier decision will never be changed, which marks the robustness of this classifier F. Hence, we formally prove Theorem 1 on the robustness bound (R) for a classifier F. □

For a deep neural network F_k, we have its robustness bound as Formula (9) shows.

$$R^k = min_{j \neq c} \frac{f_c^k(\mathbf{x}_0) - f_j^k(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(f_c^k(\mathbf{x}) - f_j^k(\mathbf{x}))||_q} \tag{9}$$

Let $g_j^k(\mathbf{x}) = f_c^k(\mathbf{x}) - f_j^k(\mathbf{x})$, we have the robustness bound as Formula (10) shows.

$$R^k = min_{j \neq c} \frac{g_j^k(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(g_j^k(\mathbf{x}))||_q} \tag{10}$$

Given S networks, combining their predictions with model averaging (*avg*), we have the ith element in the combined probability vector as $f_i^{avg}(\mathbf{x}) = \frac{1}{S}\sum_{k=1}^{S} f_i^k(\mathbf{x})$ corresponding to the robustness bound as the following Formula (11) shows.

$$\begin{aligned} R^{avg} &= min_{j \neq c} \frac{f_c^{avg}(\mathbf{x}_0) - f_j^{avg}(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(f_c^{avg}(\mathbf{x}) - f_j^{avg}(\mathbf{x}))||_q} \\ &= min_{j \neq c} \frac{g_c^{avg}(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(g_c^{avg}(\mathbf{x}))||_q} \end{aligned} \tag{11}$$

Assume the minimum of the robustness bound can be achieved with the prediction result c and j for each model F^k including the ensemble F^{avg} as Formula (12) shows.

$$\begin{aligned} R^k &= \frac{g_j^k(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(g_j^k(\mathbf{x}))||_q} \\ R^{avg} &= \frac{g_j^{avg}(\mathbf{x}_0)}{max_{\mathbf{x}} ||\nabla(g_j^{avg}(\mathbf{x}))||_q} \end{aligned} \tag{12}$$

where $g_j^k(\mathbf{x}) = f_c^k(\mathbf{x}) - f_j^k(\mathbf{x})$ and $g_j^{avg}(\mathbf{x}) = \frac{1}{S}\sum_{k=1}^{S} g_j^k(\mathbf{x})$.

We can then derive Theorem 2 that states $\exists\, k, 1 \leq k \leq S, R^k \leq R^{avg}$, where the equal sign corresponds to the case that all member networks are the perfect duplicates. This theorem further indicates that the ensembles of high diversity can improve the robustness of individual member networks.

Theorem 2 (Ensemble Robustness Bound Enhancement)**.** Let R^{avg} denote the robustness bound for an ensemble, which combines its member network predictions through model averaging (*avg*). We can always find one member network F^k with its robustness bound R^k satisfying $R^k \leq R^{avg}$.

Proof of Theorem 2. We prove the ensemble robustness bound enhancement by contradiction. First, we assume $\forall\, k, 1 \leq k \leq S, R^k > R^{avg}$, that is as Formula (13) shows.

$$g_j^k(\mathbf{x}_0)(max_{\mathbf{x}} ||\nabla(g_j^{avg}(\mathbf{x}))||_q) > g_j^{avg}(\mathbf{x}_0)(max_{\mathbf{x}} ||\nabla(g_j^k(\mathbf{x}))||_q) \tag{13}$$

For each $k \in \{1, ..., S\}$, this inequality holds. To add them all, we have Formula (14).

$$\sum_{k=1}^{S} g_j^k(\mathbf{x}_0)(max_{\mathbf{x}} ||\nabla(g_j^{avg}(\mathbf{x}))||_q) > \sum_{k=1}^{S} g_j^{avg}(\mathbf{x}_0)(max_{\mathbf{x}} ||\nabla(g_j^k(\mathbf{x}))||_q) \tag{14}$$

That is Formula (15).

$$(max_{\mathbf{x}} ||\nabla(g_j^{avg}(\mathbf{x}))||_q) \sum_{k=1}^{S} g_j^k(\mathbf{x}_0) > g_j^{avg}(\mathbf{x}_0) \sum_{k=1}^{S} (max_{\mathbf{x}} ||\nabla(g_j^k(\mathbf{x}))||_q) \tag{15}$$

Given $g_j^{avg}(\mathbf{x}) = \frac{1}{S}\sum_{k=1}^{S} g_j^k(\mathbf{x})$, we have Formula (16):

$$(max_{\mathbf{x}} ||\nabla(\sum_{k=1}^{S} g_j^k(\mathbf{x}))||_q) \frac{1}{S} \sum_{k=1}^{S} g_j^k(\mathbf{x}_0) > \frac{1}{S} \sum_{k=1}^{S} g_j^k(\mathbf{x}_0) \sum_{k=1}^{S} (max_{\mathbf{x}} ||\nabla(g_j^k(\mathbf{x}))||_q) \tag{16}$$